Phase 1 Pancreatic Cancer Trial: Efficient LLM Document Guidance

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Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. The DOI placeholder `10.5281/zenodo.21018646` is filled at deposit. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference.

Abstract

A Phase 1 oncology trial advances only when a sequence of large regulatory and clinical documents is authored, reviewed, and approved, while patients wait for that paperwork to be assembled. This paper illustrates a repository based large language model (LLM), driven by a single master prompt that writes and executes a schedule of sub-prompts committed to GitHub in real time; which can hasten the entire Phase 1 process by generating documents through an auditable "mermaid" to "draft" to "full" to "final" pipeline. The method was conducted on a prior pancreatic ductal adenocarcinoma program: an on-premises LLM directed robotic Whipple with perioperative daraxonrasib (RMC-6236), whose Phase 1 and Phase 2 protocols were themselves single-prompt multi-file builds. The following trial aspects were addressed: the initial IND and IRB package; protocol amendments with synchronized consent; cohort-review packages; the complete clinical-hold response; the Phase 2-to-3 briefing package; and the CSR and NDA/BLA modules. Only the administrative and preparation time bucket compresses, while clinical follow-up and fixed regulatory review clocks do not. Papers finish in 1-4 days, based on prompt and input status. Twenty-four colored Mermaid converted figures ground the prose in LaTeX. Paper repository, directory, and file name samples were verified manually by the author. For terminal patients, probable benefit exceeds probable risk.

Keywords Pancreatic ductal adenocarcinoma; Phase 1 clinical trial; document generation; repository based LLM; sub-prompt schedule; clinical trial acceleration; daraxonrasib.

1 Introduction

A Phase 1 oncology trial is, from a documentation standpoint, a relay of large written artifacts. Before a single patient is dosed, an Investigational New Drug (IND) dossier, an Investigator's Brochure, a clinical trial protocol, and an informed consent form must be authored and approved; during the trial, safety narratives, amendments, and cohort-review packages must be produced on short clocks; and after the trial, a clinical study report and downstream regulatory modules must be assembled. Each of these documents can sit on the critical path of the next decision. This paper (paper v1.0, within repository v4.2.0) shows that a single repository based large language model (LLM) prompt, which first writes and then executes a schedule of sub-prompts, can generate every one of these documents through a mermaid to draft to full to final pipeline, and can do so in a way that is monitorable and verifiable in real time.

1.1 Patients Currently Enrolled in Trial Cannot Wait for Slowdowns

Pancreatic ductal adenocarcinoma (PDAC) has a five-year overall survival below 13 percent and is among the most lethal solid tumors, so a patient already enrolled in a Phase 1 PDAC study has little survival margin to spare on administrative delay [1, 2]. The clinical and operational time of a trial (recruitment, treatment, and waiting for progression-free and overall survival to mature) cannot be compressed by writing faster, and neither can the fixed regulatory review clocks. What can be compressed is the administrative and preparation time spent authoring the large documents between steps. Faster paperwork therefore means faster treatment: an internally consistent package can reach the IRB and the FDA sooner, sites can activate sooner, and the next cohort can open sooner. In this work, a new iteration of any large document (a fresh draft, a full revision, or a final polish) completes in one to four days rather than the weeks to months a sequential human authoring cycle typically consumes. Figure 1 shows the single-prompt pipeline that produces these iterations, and Figure 2 shows the one-to-four-day cadence against the traditional baseline.

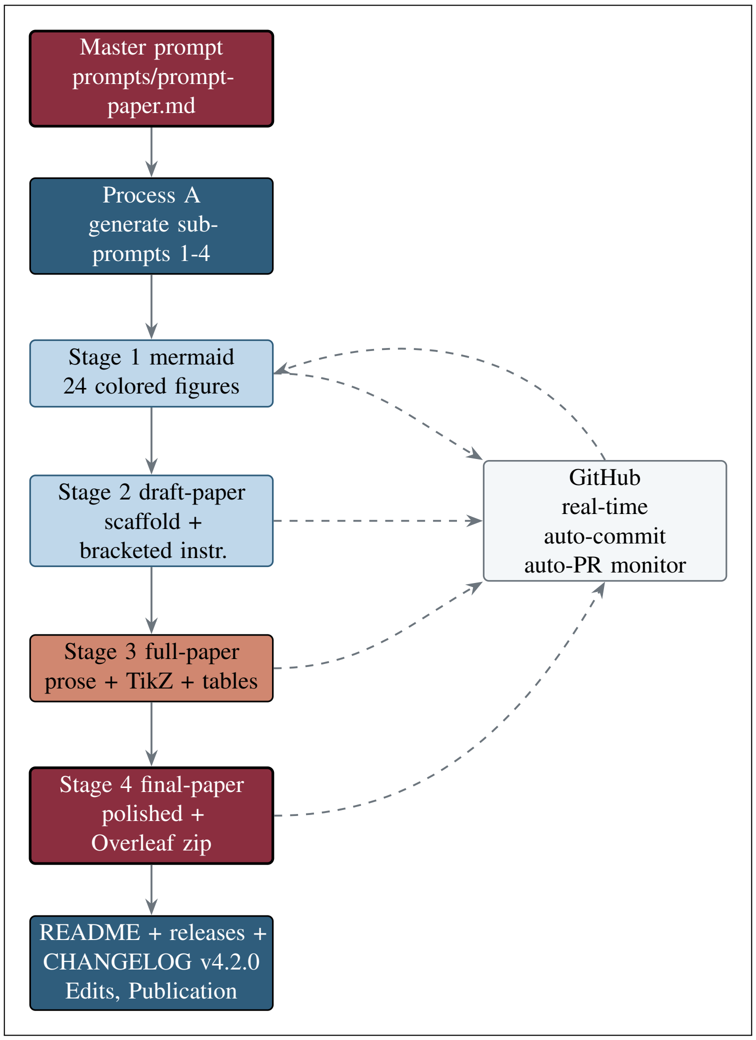

Figure 1. The single-prompt "mermaid" to "draft" to "full" to "final" build pipeline. One master prompt drives Process A (generate the four sub-prompts) and then Process B (execute them in order). Dashed edges show that every stage auto-commits and opens a pull request on GitHub in real time, and the curved feedback edge shows author review returning into the build, all without manual intervention.

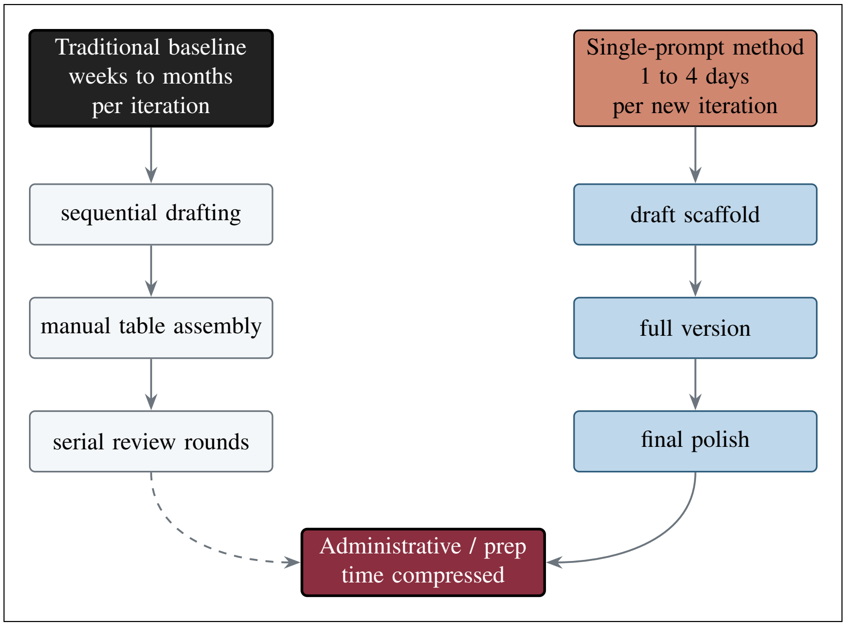

Figure 2. The traditional baseline (left) spends weeks to months per large-document iteration in sequential drafting, manual table assembly, and serial review. The single-prompt method (right) produces a draft, a full version, and a final polish in one to four days, compressing the administrative and preparation bucket only.

1.2 Effective Verifications Prove Utility

Speed is worthless if it cannot be trusted, so the method is built to be verified rather than asserted. Five checks make the build self-evidently grounded. First, the prose is grounded in 24 colored Mermaid figures that carry the real quantitative content (document counts, reporting windows, and review clocks) rather than decoration. Second, each figure matches the surrounding paper context and is cited where it is discussed. Third, every generated file is committed to GitHub in real time, so the author can view each artifact as it appears on the branch. Fourth, the paper's repository and directory names match the repository names exactly, so a reader can navigate from a sentence to the directory it describes. Fifth, the paper's file names match the repository file names, so a cited `sections/sec-04-results.tex` is the file on disk. These five checks are collected in the Results.

1.3 Patient Lives Extended: Probable Benefit Exceeds Probable Risk

The motivation is ultimately clinical. The probable benefits of the method are faster paperwork, faster treatment, a compressed administrative and preparation bucket, and more figures available for human oversight. The probable risks are the familiar LLM failure modes: an occasional generation error, a formatting artifact, or an unverified claim. Each risk is reduced to a low residual by a process control: human-in-the-loop review of commits, Mermaid grounding, paper-to-repository name matching, real-time GitHub monitorability, and a senior-author proofreading pass. Figure 3 weighs the two sides. For a population whose disease will not wait, withholding a method that safely compresses preparation time carries its own cost, and the balance favors using it.

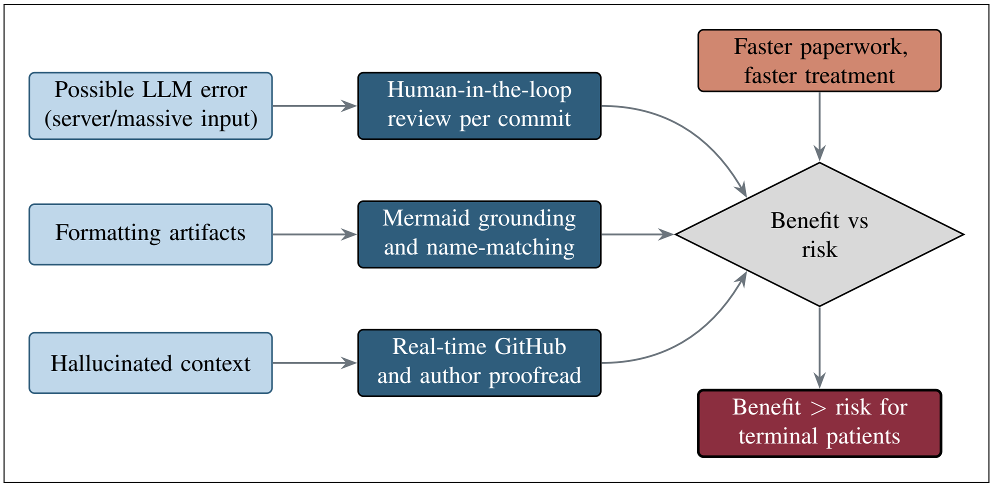

Figure 3. Each probable risk (left) is reduced to a low residual by a process control (center); the decision node (right) integrates the mitigated risks with the weighted benefits and resolves in favor of terminal patients.

1.4 Scope, Contributions, and Relationship to the Adoption Guide

This paper builds directly on the *Oncology Trial PI LLM Adoption Guide* [3], which it uses both as the paper template and as the practical hands-on guide whose Repository Setup, Prompt Proficiency, Project Proficiency, and Figure and Table guidance it turns into a worked Phase 1 pancreatic cancer document-generation study.

The contributions are fourfold: (1) a single-prompt mermaid to draft to full to final pipeline that produces a complete scientific paper and, by the same mechanism, the large Phase 1 trial documents; (2) explicit coverage of the six document targets where faster authoring yields the greatest schedule value; (3) twenty-four new, professionally colored figures that reproduce identically from Python Mermaid to LaTeX TikZ; and (4) a real-time, monitorable, name-matched build. The demonstration is anchored to a real-world program: the on-premises LLM directed robotic Whipple with perioperative daraxonrasib (RMC-6236), whose Phase 1 [4] and Phase 2 [5] protocols and whose 60-second simulation [6] were themselves single-prompt multi-file builds, and to the National Platform for Physical AI Oncology Trials as the governing framework [2].

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2 Methods

2.1 Repository Based LLM and Input Curation

The method places a finite set of appropriately sized, high-quality inputs in a single repository directory so that a 1M-context LLM can process them efficiently [3]. The inputs are the paper template (`inputs/llm-adoption`), the bibliography (`inputs/references.bib`), and the four research sources (`research/document-types` and `research/industry-workflow`), each pre-chunked with its own README so the next processing step executes optimally. Two practical limits are respected: total input is kept under 5 MB (0.84 MB prior to publication files), and individual README files are kept under roughly 25K tokens to avoid per-file errors. Outputs are generated in sections (one `.tex` per paper section) so that text and code form right-sized chunks for downstream reuse. Figure 4 shows architecture.

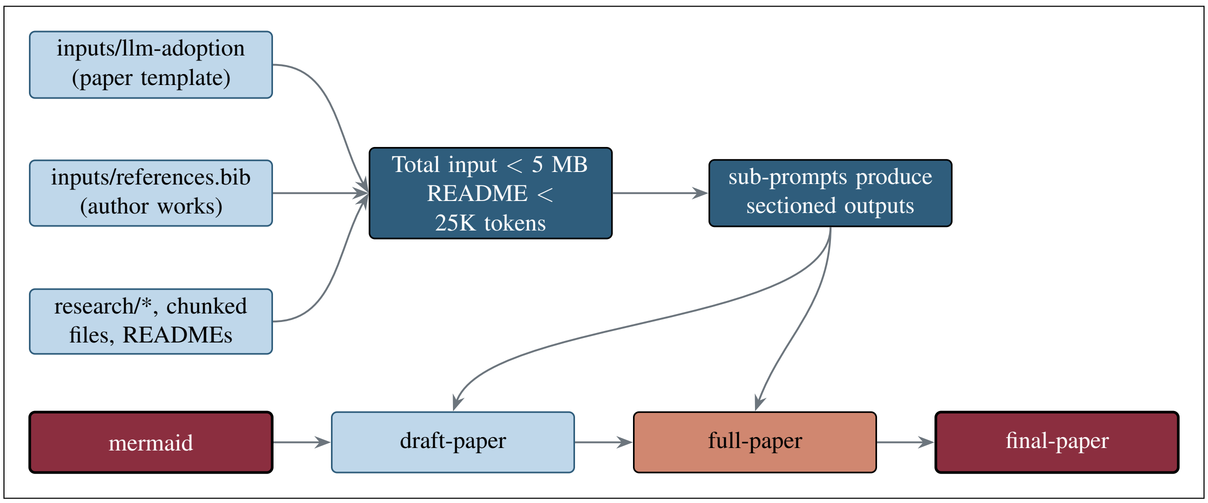

Figure 4. Curated inputs feed a 1M-context model under two size limits; sub-prompts produce sectioned outputs that become the four stage directories, each reusable as a right-sized chunk for future builds.

2.2 Single Prompt, Sub-Prompt Schedule, and Stages

A single master prompt (`prompts/prompt-paper.md`) drives the whole build. *Process A* reads that prompt and writes the four stage sub-prompts in `sub-prompts/`; *Process B* then executes them in order, growing the work from Mermaid figures to a draft, a full, and a final LaTeX paper, with outputs accumulating as project-scale context for the next stage [3]. The schedule is adapted from the trial-protocol build (`trial-protocol/sub-prompts`). Figure 5 shows the generate-then-execute structure.

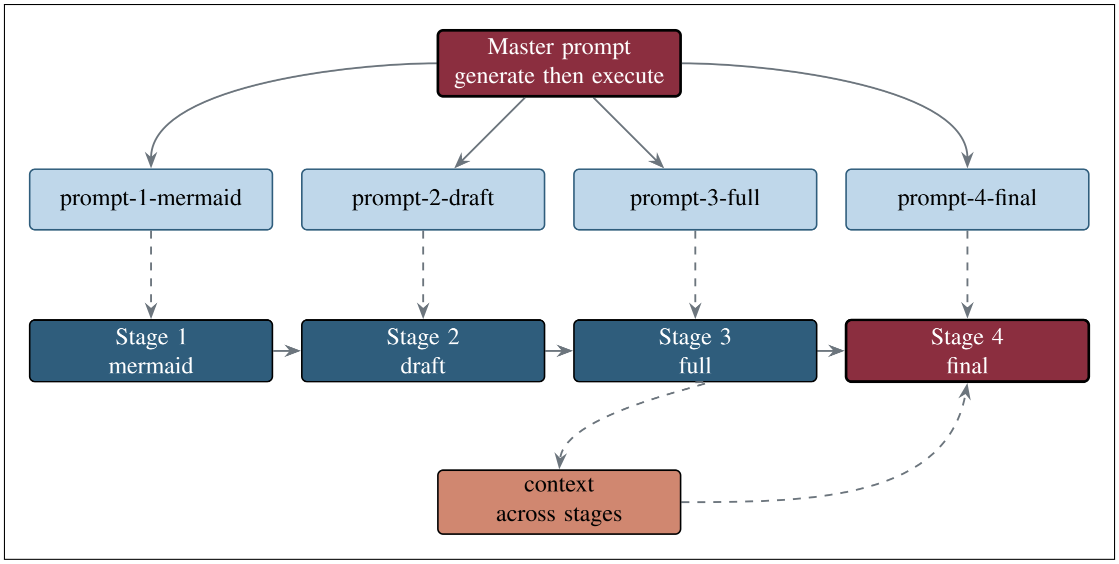

Figure 5. Process A generates the four sub-prompts; Process B executes them sequentially as Stages 1 to 4. Outputs accumulate as context that informs the next stage, so the final paper is produced under one prompt.

2.3 The Phase 1 Document Landscape

Phase 1 oncology trials depend on large documents before, during, and after the study [7, 8]. Before the trial: the initial IND dossier, the Investigator's Brochure, the clinical trial protocol, and the informed consent form (often 20 or more pages, written to a 6th-to-8th grade reading level). During the trial: serious adverse event and SUSAR safety narratives on 7-day and 15-day clocks, the annual Development Safety Update Report, protocol amendments, and dose-escalation meeting minutes. After the trial: tables, listings, and figures, the clinical study report under ICH E3, manuscripts and congress abstracts, and lay summaries. Figure 6 maps this landscape.

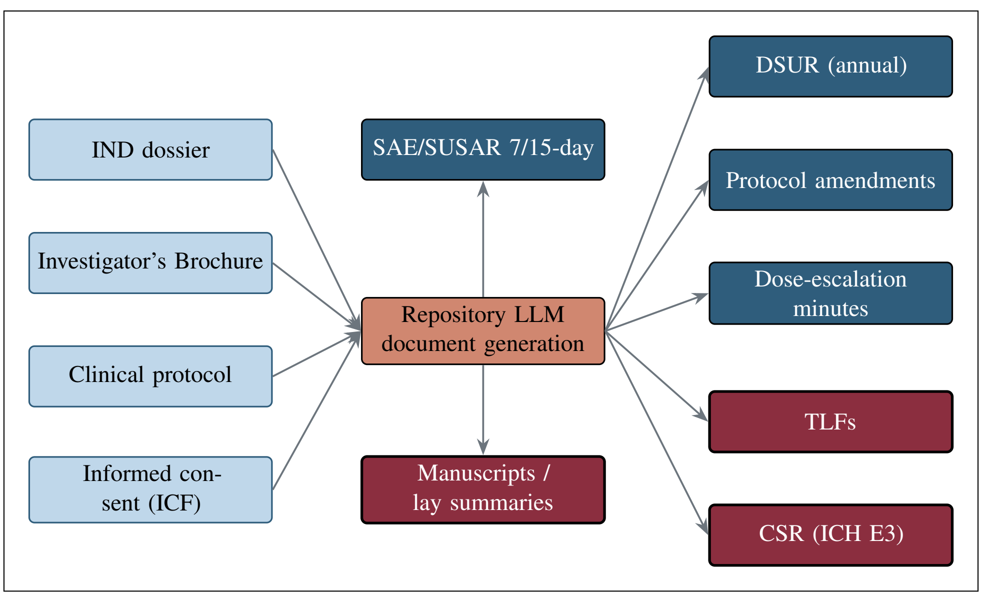

Figure 6. The large Phase 1 oncology document landscape, grouped into before-trial inputs (left), during-trial process documents (upper right), and after-trial outputs (lower right), all generated by the repository LLM.

2.4 Document Decision-Gate Taxonomy

Whether faster authoring speeds a trial depends on the kind of gate a document controls [8]. A hard gate means the trial legally or ethically cannot proceed; a protocol-defined gate means the trial's own rules block the next cohort, arm, or adaptation; and a decision gate means progress is legally possible but the sponsor will not invest without an internal or regulatory decision. Faster creation speeds the trial only when the document is on the critical path. Figure 7 shows taxonomy and Table 1 states it in full.

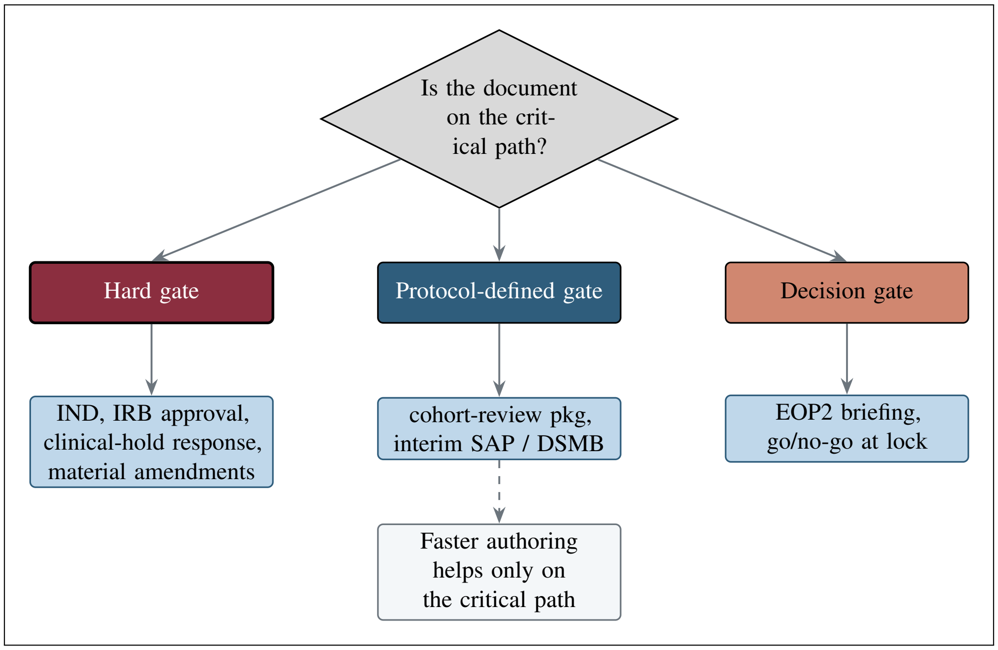

Figure 7. The hard / protocol-defined / decision gate taxonomy. The schedule value of faster authoring is realized only for documents on the critical path.

Gate type	Meaning	Example documents and effect of faster authoring
Hard gate	The trial legally or ethically cannot proceed	Initial IND, IRB approval, clinical-hold response, and material amendments; faster authoring starts the review clock and all sites sooner but cannot shorten a fixed review period.
Protocol gate	The trial's own rules block the next cohort, arm, or adaptation	Cohort-review packages and the interim-analysis statistical plan or DSMB charter; faster authoring shortens the pause after the data mature but not the observation window itself.
Decision gate	Legally possible, but the sponsor will not invest without a decision	The End-of-Phase 2 briefing book and the go/no-go assessment after database lock; faster authoring lets the sponsor request the meeting and begin scheduling sooner.

Table 1. The three document decision-gate types, their meaning, and the effect of faster authoring (adapted from the document-types research sources).

2.5 Before, During, and After Phase 1: Data and Document Workflow

Data collection in modern oncology trials runs through a digital ecosystem [7]. Before the trial, clinical research coordinators extract data from electronic health records into an electronic data capture (EDC) system and record baseline tumor measurements by RECIST. During the trial, electronic case report forms capture dose, pharmacokinetic, and pharmacodynamic data; electronic clinical outcome and patient-reported outcome devices capture symptoms; and adverse events are graded by CTCAE, with dose-limiting toxicities captured in the 3+3 window. After the trial, data undergo source document verification by clinical research associates before the database is locked. Figure 8 shows the pipeline.

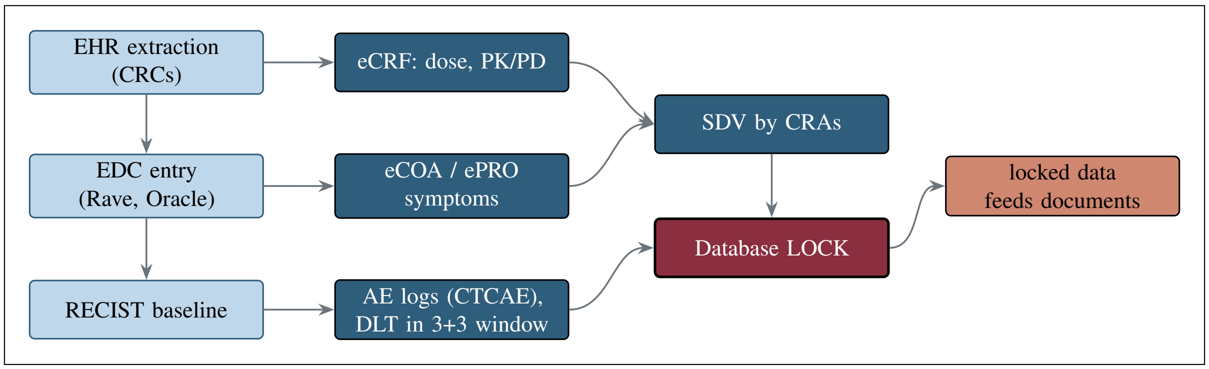

Figure 8. The before / during / after data-collection pipeline, from EHR extraction through EDC, eCRF, eCOA/ePRO, and CTCAE-graded adverse events to source document verification and database lock, after which the locked data feed document generation.

Medical writers act as the project managers of document creation, working in an electronic document management system such as Veeva Vault with controlled, version-tracked authoring and standardized templates [9, 7]. Figures 9, 10, and 11 show the before, during, and after authoring flows respectively.

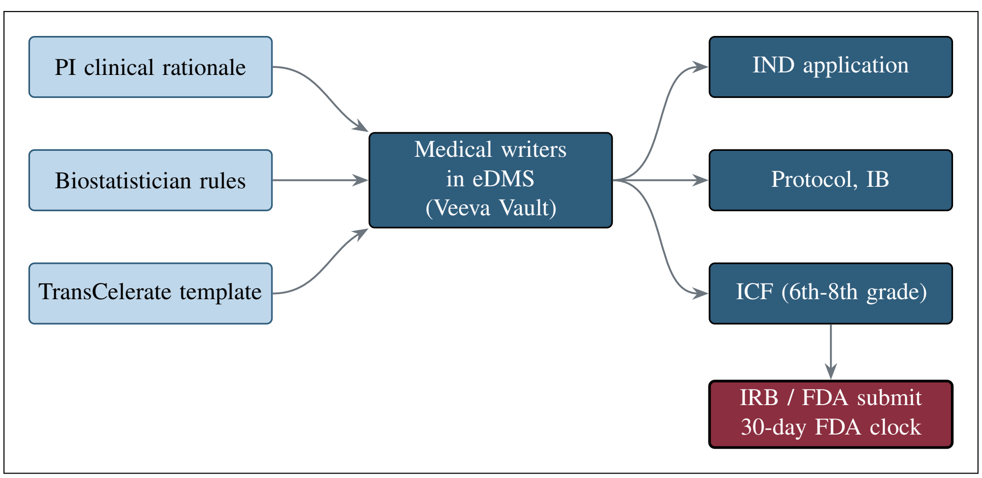

Figure 9. Pre-trial authoring. PI, biostatistician, and template inputs are compiled by medical writers in an eDMS into the IND, protocol, Investigator's Brochure, and informed consent form, which start the 30-day FDA clock on submission.

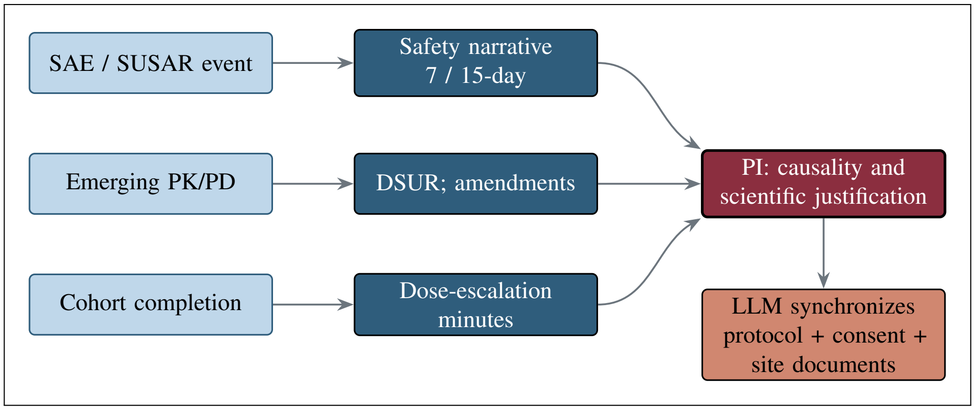

Figure 10. During-trial authoring is driven by safety data: narratives on 7-day and 15-day clocks, the DSUR, amendments, and escalation minutes, with the PI assessing causality and the LLM synchronizing the protocol, consent, and site documents.

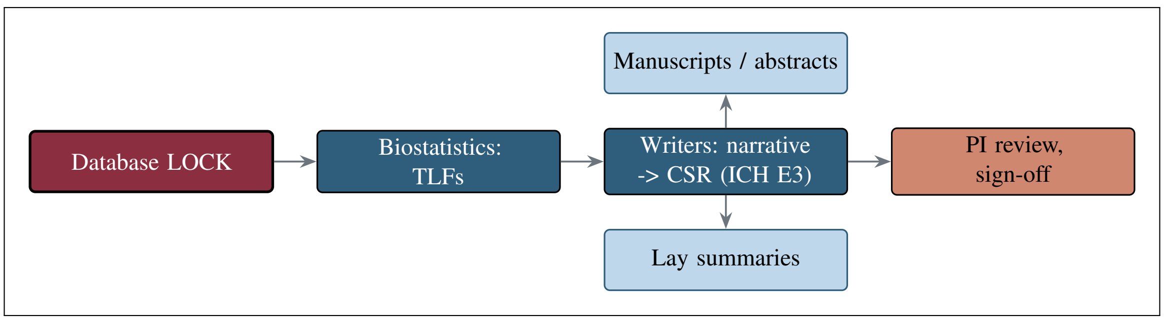

Figure 11. After-trial authoring: from database lock, biostatisticians generate tables, listings, and figures; writers draft the narrative into the CSR under ICH E3; the PI reviews and signs off; and manuscripts and lay summaries branch off.

2.6 Figures, Tables, and Formatting Method

All figures use one five-step palette so that emphasis reads by fill and stroke weight alone and each Mermaid figure reproduces 1:1 as a TikZ `mermaidfig` (Figure 12). Body text stays black, as in the paper template; only figures carry color. Tables are set to the body-text width with ragged-right (`>{\raggedright\arraybackslash}`) columns so no cell shows large interword gaps, following the formatting strategies in `trial-protocol/final-protocol/publication`. Paper section architecture maps each section to a `.tex` file (Figure 14), and figure grounding with Mermaid sources (Figure 15).

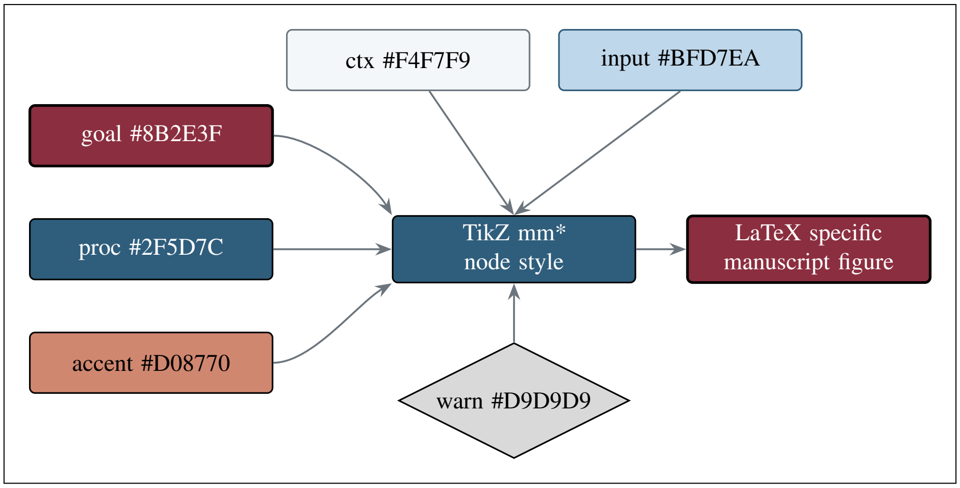

Figure 12. The five-color palette (plus the gray decision fill) maps 1:1 from the Mermaid `classDef` classes to the TikZ `mm*` node styles, so every figure looks the same in LaTeX as on GitHub.

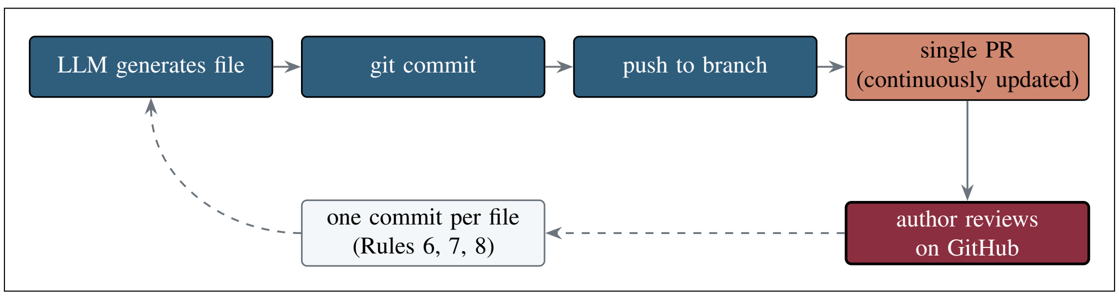

Figure 13. The real-time auto-commit and auto-PR loop. Each generated file is committed and pushed at once into one continuously updated pull request that the author monitors, with one commit per file per the commit rules.

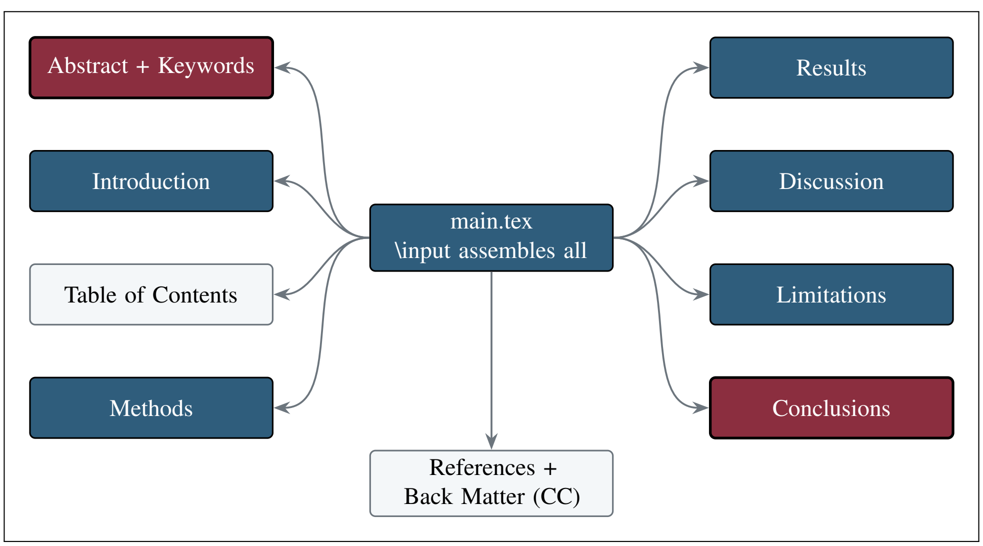

Figure 14. The paper section architecture. Each PAPER FORMAT section is one `sections/*.tex` file assembled by `main.tex`; the Table of Contents is generated after the Introduction.

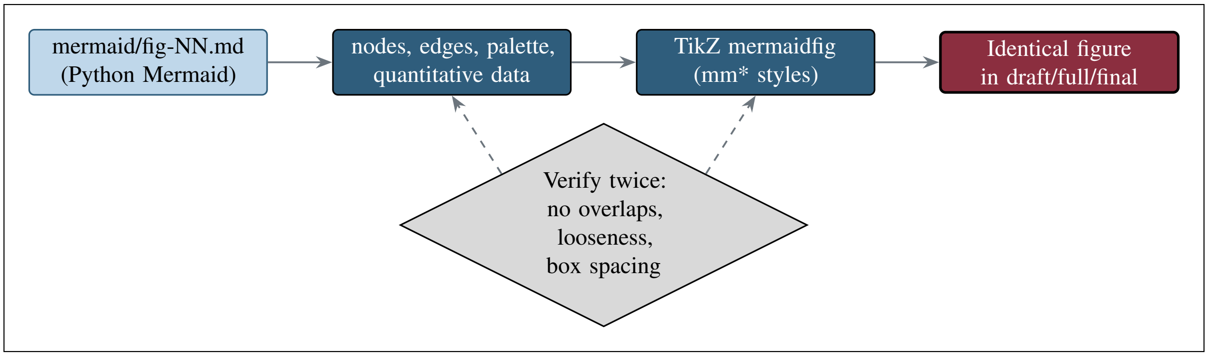

Figure 15. Figure grounding: each Mermaid source is reproduced as a TikZ `mermaidfig` of the same complexity, then verified twice for overlaps, arrow looseness, and box spacing.

2.7 Real-Time Auto-Commit and Auto-PR Monitorability

To keep the human in the loop, the build commits each file to GitHub the moment it is generated and opens a single continuously updated pull request, so the author can review artifacts as they appear without intervention (Figure 13). Commit discipline follows the rules: one commit for each of `main.tex`, the style file, the bibliography, and the README, and one commit per section `.tex`; the second-to-last commit of each stage fixes all errors, and the last performs the remaining repository updates.

3 Results

3.1 A Single Prompt Produced Four Stage Outputs

Executing the single master prompt produced four stage outputs in `trial-documents`: 24 colored Mermaid figures, a draft scaffold, a full paper, and a final paper. Each LaTeX stage ships `main.tex`, `paperstyle.sty`, `references.bib`, a `sections/` directory of eight section files, and an Overleaf zip, and every distinguishable file was a commit pushed in real time. Table 2 summarizes stage outputs.

Stage	Directory	Artifacts	Commits
1 Mermaid	`mermaid/`	24 colored Mermaid figure files, README, output-mermaid.md	26
2 Draft	`draft-paper/`	main.tex, paperstyle.sty, references.bib, 8 section scaffolds, README, zip	10+
3 Full	`full-paper/`	main.tex, style, bib, 8 full sections with 24 TikZ figures and tables, README, zip	10+
4 Final	`final-paper/`	polished main, style, bib, 8 sections, README, zip; root repository updates	10+

Table 2. The four stage outputs produced by the single prompt, each with one commit per file set pushed in real time.

3.2 The Six Acceleration Targets Addressed

The six document targets where faster, validated authoring yields the greatest schedule value were each addressed by the method, with the binding clock identified for each (Figure 16, Table 3). Node color in Figure 16 encodes the gate type from Figure 7: maroon for hard gates, steel blue for protocol-defined gates, and terracotta for decision gates. Two targets are detailed further: the initial IND and IRB package, the single highest-value target (Figure 17), and the complete clinical-hold response, whose 30-day FDA review begins only when a complete response to every deficiency is received (Figure 18).

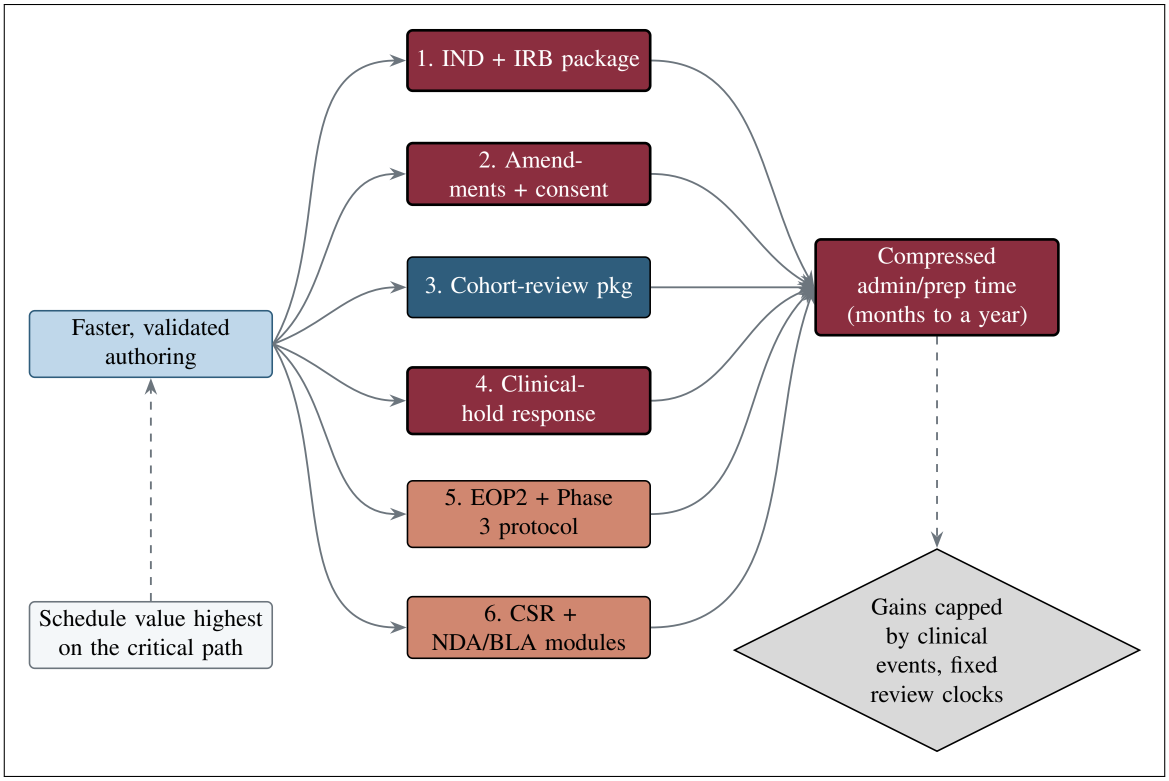

Figure 16. The six greatest-acceleration document targets. One input, faster validated authoring, feeds all six ranked targets, each contributing to a common compression of administrative and preparation time; the schedule value is highest on the critical path, and the gains are capped by clinical events and fixed review clocks.

#	Document target	Gate	Binding clock or constraint
1	Initial IND and IRB package	Hard	30-day FDA review; IRB meeting calendar; cannot generate missing tox or stability data
2	Protocol amendments + synchronized consent/site	Hard	IRB approval; speed is lost if protocol, consent, and database are revised serially
3	Cohort-review packages after safety data mature	Protocol-defined	The protocol-required DLT observation window
4	Complete clinical-hold response	Hard	30-day FDA review of a complete response to every deficiency
5	Phase 2-to-3 briefing package and Phase 3 protocol	Decision	EOP2 meeting scheduling and review
6	Pivotal CSR and NDA/BLA modules after database lock	Decision/filing	Database lock; query resolution; RTOR staging

Table 3. The six acceleration targets, their gate type, and the binding clock that faster authoring cannot shorten (adapted from the document-types sources).

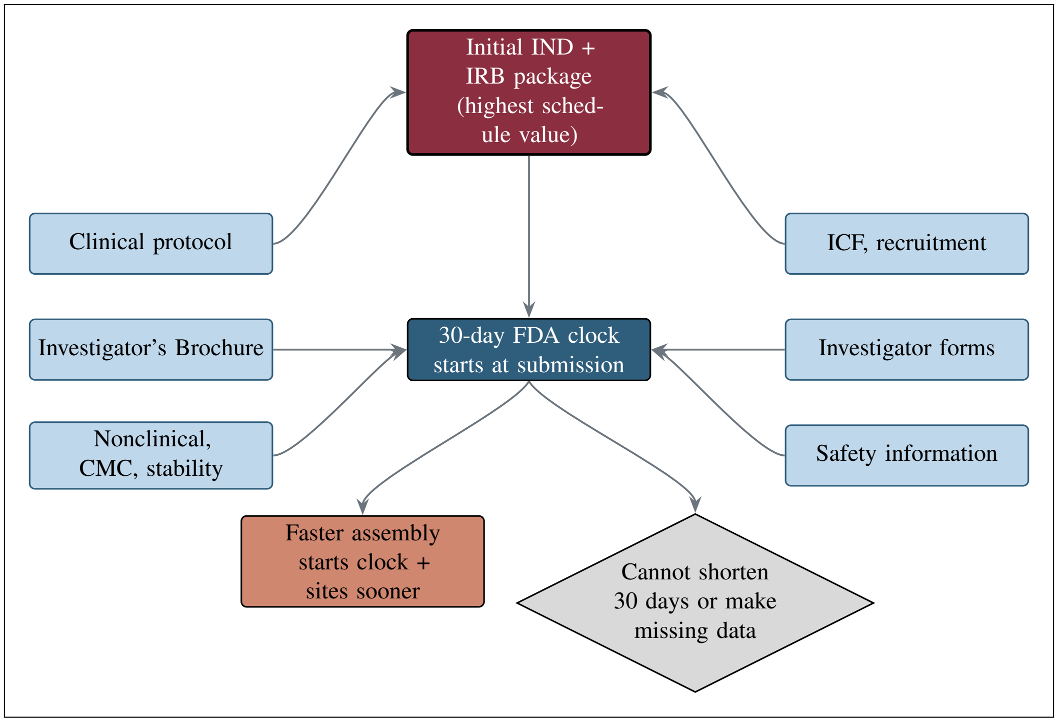

Figure 17. Composition of the initial IND and IRB package. Faster, internally consistent assembly starts the 30-day FDA clock and all sites sooner but cannot shorten the fixed period or generate missing toxicology or stability data.

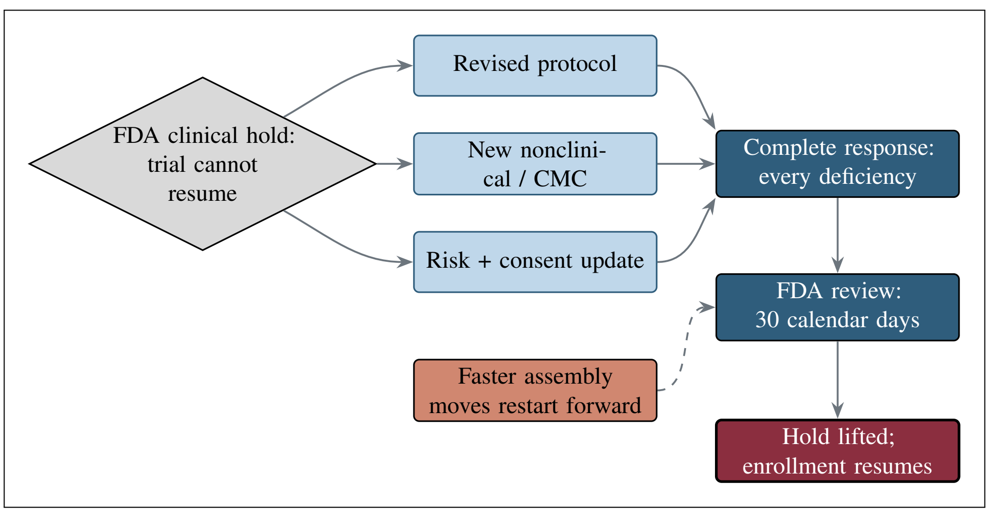

Figure 18. The 30-day FDA review begins only when a complete response to every deficiency is received, so faster assembly moves the potential restart date forward; an incomplete response does not start the useful clock.

3.3 New Document Iterations in One to Four Days

A new iteration of any large document completed in one to four days, against a weeks-to-months baseline. The gain is confined to the administrative and preparation bucket: the clinical and operational bucket (waiting for survival endpoints to mature) and the fixed regulatory review bucket are unchanged (Figure 19). Cumulatively, compressing only the writing bucket can save months, and sometimes more than a year, across a program [10].

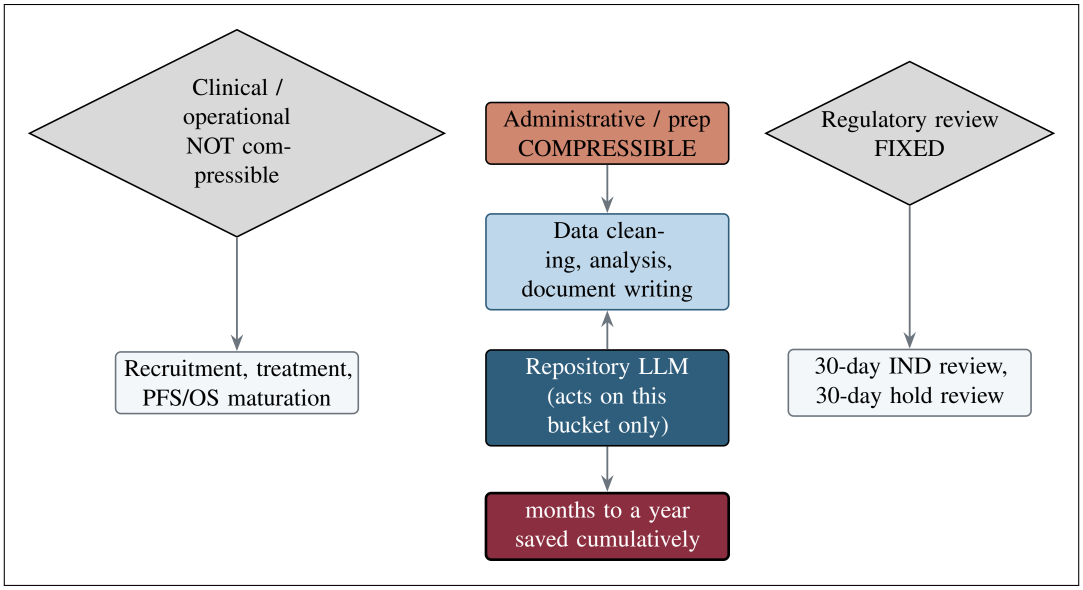

Figure 19. The three timeline buckets. Only the administrative and preparation bucket compresses; clinical operational bucket and fixed regulatory review clocks do not. Repository LLM acts on middle bucket only.

3.4 Effective Verifications: Grounding and Name-Matching

Five checks make the build verifiable (Figure 20, Table 4): grounding via the 24 Mermaid figures; figures matching the surrounding context; real-time GitHub visibility of each file; repository and directory name matching; and file-name matching. Quality gates that prevent a document from generating a rework cycle (amendments, regulator questions, IRB revisions, or extra review cycle) are in Figure 21.

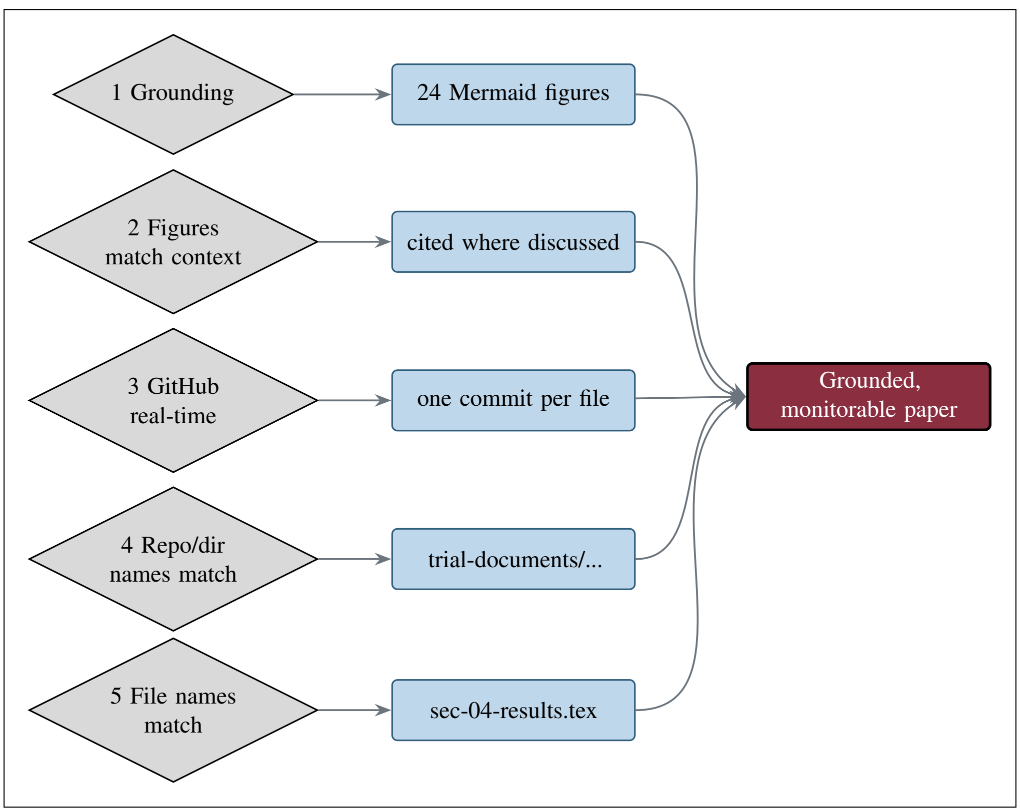

Figure 20. Five name-matching and grounding verifications, each with a concrete example.

#	**Verification check**	**Concrete example**
1	Grounding via Mermaid figures	The 24 figures in `mermaid/` carry the document counts and review clocks reused in the prose
2	Figures match paper context	Each figure is cited in the sentence that discusses it
3	AI files viewable on GitHub in real time	One commit per file is pushed at generation, so each artifact appears on the branch as it is made
4	Repository and directory names match	The paper refers to `trial-documents/mermaid` and `full-paper/sections`, which are the directories on disk
5	File names match repo file names	A cited `sec-04-results.tex` is this file; `fig-04-acceleration-targets.md` is the figure file

Table 4. The five verification checks (OUTLINE theme 2) with a concrete example for each.

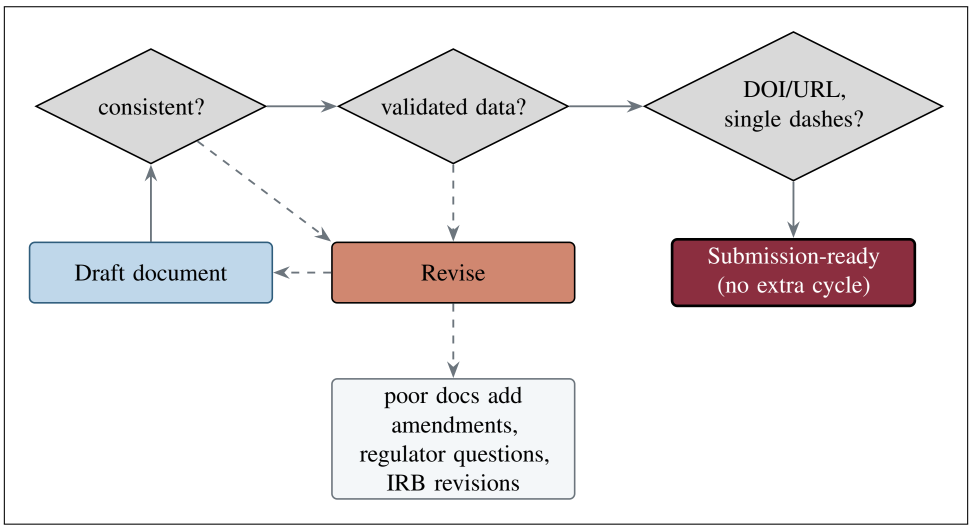

Figure 21. A draft that fails consistency, data support, or formatting checks is revised rather than submitted, preventing amendments, regulator questions, IRB revisions, or extra review cycle that inconsistent documents cause.

3.5 Prior Single-Prompt Multi-File Developments as Evidence

This project is the latest in a line of single-prompt, multi-file builds. The Phase 2 protocol directory [5] is the most directly analogous: like this paper, it was grown from Mermaid figures through draft, full, and final LaTeX under one prompt. The law and bill repositories did the same for legislative text: the review [11], adoption [3], and enactment [12] directories, and the two auto-bill directories [13, 14]. Table 5 lists them.

Repository / directory	Output type	DOI
`trial-phase-2`	Phase 2 protocol (mermaid/draft/full/final)	10.5281/zenodo.20807027
`law-physical-ai-trials/review`	Narrative case for verified trials	10.5281/zenodo.20685379
`law-physical-ai-trials/adoption`	Oncology Trial PI LLM Adoption Guide	10.5281/zenodo.20843290
`law-physical-ai-trials/enactment`	Framework for enacting H. R. 9510	10.5281/zenodo.20726461
`single-prompt-bill/auto-bill-02`	H. R. 9510 Bill v5.0	10.5281/zenodo.20619762
`single-prompt-bill/auto-bill-01`	H. R. 9510 Bill v4.0	10.5281/zenodo.20576907

Table 5. Prior single-prompt multi-file developments cited as evidence for the analogous approach used here.

3.6 2025 Application Papers Establishing LLM Trust

LLM trust for this application was established across 2024 to 2026 (Figure 22, Table 6). The 2025 application papers progressed from drug-cost reasoning and clinical decision support [15, 16, 17] through glioblastoma meta-analysis [18], lung adenocarcinoma [19], PDAC digital-twin proposals [1], a 100,000-patient in silico Phase III [20], QSP simulation [21], FDA digital-twin compliance [22], end-to-end oncology-trial LLM efficiency [23], glioblastoma drug-synergy and patient-matching machine learning [24, 25], and a 16-LLM code-generation competition [26].

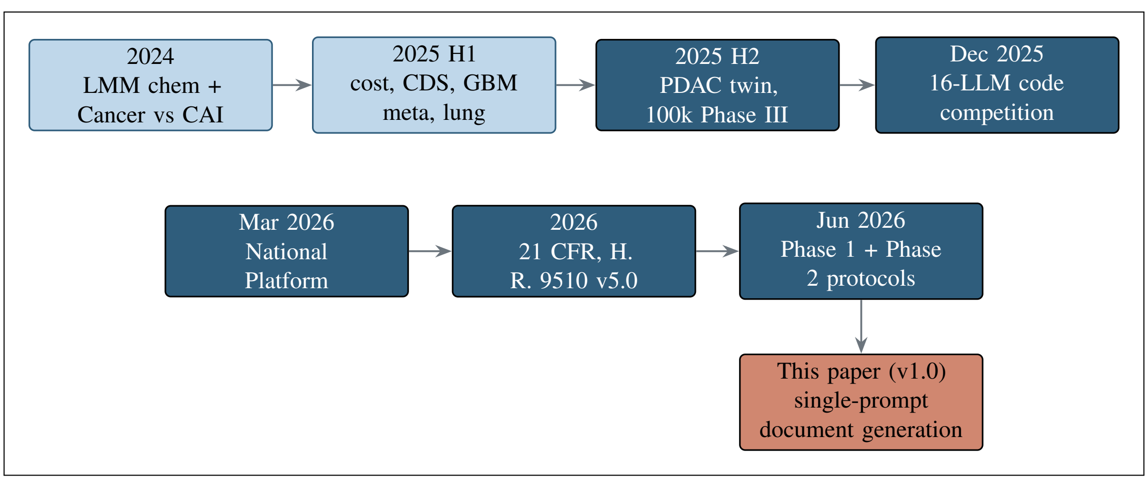

Figure 22. The author LLM-trust evidence timeline from 2024 to 2026, terminating in this single-prompt document-generation paper.

Date	Work	Contribution to trust
2025-06	PDAC digital-twin proposals [1]	End-to-end PDAC trial proposal scaffolding
2025-07	100,000-patient in silico Phase III [20]	Large-scale simulated trial generation
2025-08	QSP metastatic PDAC simulation [21]	Code, VVUQ, playbook: protocol to predict
2025-10	FDA digital-twin compliance; oncology LLM efficiency [22, 23]	Compliance-aware, efficient document workflows
2025-11/12	GBM drug synergy; AI peer review; 16-LLM competition [24, 25, 26]	Verified, comparative LLM code generation

Table 6. Representative 2025 application papers and their contribution to the LLM trust on which this project builds.

4 Discussion

4.1 Probable Benefit Exceeds Probable Risk

The central claim is that, for a Phase 1 PDAC population, the probable benefit of repository based LLM document generation exceeds its probable risk. The benefits are concrete and on the critical path: faster paperwork, faster treatment, a compressed administrative and preparation bucket, and a larger number of figures available for human oversight than a manual process would produce. The risks are the known LLM failure modes, and each is bounded by a specific control rather than left to trust. An occasional generation or server error is caught by human-in-the-loop review at every commit; a formatting artifact is caught because each figure is grounded in a Mermaid source and each file is visible on GitHub; an unverified claim is caught by the name-matching and proofreading passes. The residual risk after these controls is low, while the benefit is realized on every iteration. For patients whose disease will not pause, the alternative (continuing to author large documents by slower sequential means) carries its own, often larger, cost in delayed treatment.

4.2 Mechanism: Compressing Administrative and Preparation Time

The mechanism is specific. A clinical trial timeline divides into clinical and operational time, administrative and preparation time, and fixed regulatory review time [10]. Writing faster does not make a tumor respond faster and does not compel a regulator to review faster; it compresses only the middle bucket. That bucket, however, contains the document white space between phases, which is often where months are lost. By generating the six highest-value documents prospectively and in synchronized form (so that protocol, consent, database, and site documents move together rather than serially), the method removes precisely the idle authoring time that separates the end of one step from the start of the next.

4.3 Prior Developments as Evidence for This Project

This paper did not arrive without precedent. The author's 2024 to 2026 works established, in stages, that LLMs can be trusted for oncology trial artifacts: from drug-cost reasoning and clinical decision support, through glioblastoma meta-analysis and lung adenocarcinoma reviews [18, 19], to PDAC digital-twin proposals and large in silico trials [1, 20, 21], to compliance-aware efficiency and verified comparative code generation [23, 26]. The National Platform for Physical AI Oncology Trials provided the governing framework [2], and the Phase 1 [4] and Phase 2 [5] protocols, together with the law and bill repositories [11, 3, 12, 13, 14], demonstrated the single-prompt multi-file build on real regulatory and legislative documents. Applying the same single-prompt approach to the document-generation problem itself is the natural next step.

4.4 Why Trust the Method: Verification, Not Faith

The method is designed around verification before generation [13]: a document is trusted because it is grounded, name-matched, monitorable, and quality- gated, not because the model is assumed correct. This matters as AI outputs grow more complex and voluminous. The adoption guide frames the underlying question politely but pointedly: as AI-directed artifacts become larger and more intricate, why would human-only review remain sufficient, rather than a discipline that makes every claim traceable to a source and every figure traceable to a Mermaid file? Figure 21 quality gates are operational answer.

4.5 Relationship to the Real-World Daraxonrasib Trial

The documents accelerated here are exactly those the real daraxonrasib PDAC program requires. Figure 23 traces the document thread from a KRAS-mutated PDAC patient and the daraxonrasib (RMC-6236) plus LLM-directed robotic Whipple intervention, through the Phase 1 combined IND/IDE protocol [4] and the 3+3 dose-escalation cohort-review packages that establish the recommended Phase 2 dose, to the amendments, the Phase 1-to-2 transition documents, and the randomized Phase 2 protocol [5, 6]. Figure 24 closes the argument: faster authoring shortens the white space between these documents, which advances treatment and, for this population, extends progression-free and overall survival.

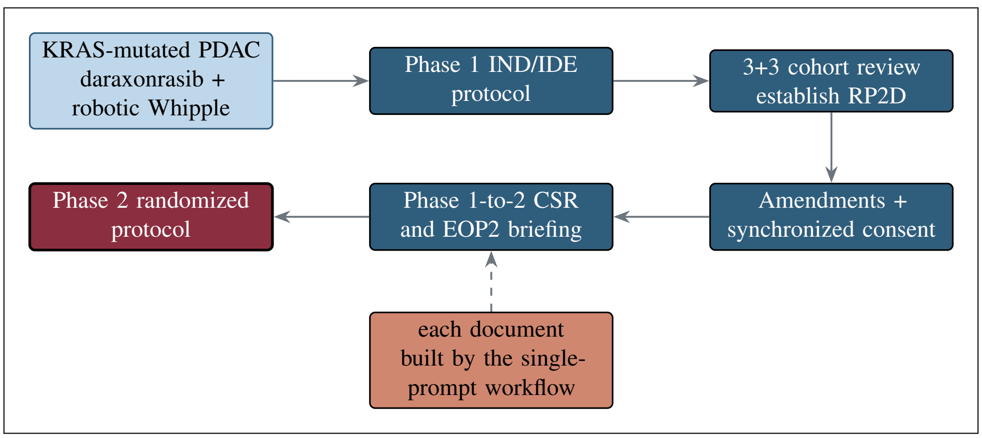

Figure 23. The real-world daraxonrasib PDAC document thread, from patient and intervention through the Phase 1 protocol, cohort-review packages, amendments, and transition documents to Phase 2 randomized protocol, each built by single-prompt workflow.

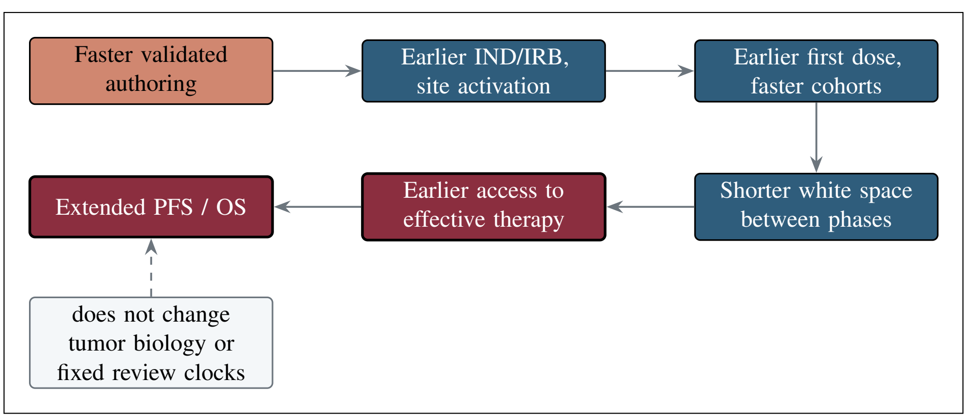

Figure 24. The patient time-saved cascade. Faster authoring advances each step from submission to first dose, shortening the white space between phases and, for this population, extending progression-free, overall survival, without changing tumor biology or fixed review clocks.

5 Limitations and Future Work

5.1 Limitations

The approach does not rely on live internet search for large projects, because specifying single external URLs is unreliable at scale; instead it depends on a curated repository of appropriately sized inputs [3]. Because the LLM does not always have access to a PDF compiler, some senior-author formatting (table column widths, figure-caption spacing, and occasional page breaks) must still be applied by hand, and the paper explicitly performs that pass in the final stage. Total input is kept under roughly 5 MB and individual files under per-file token limits, which bounds how much source material a single build can absorb. High-dimensional or approximate inputs can make outputs less deterministic, and oversized or overly complex inputs can surface server errors that signal reduced quality. Most fundamentally, the method changes none of the clinical realities: it does not alter tumor biology, shorten dose-limiting toxicity windows or survival follow-up, or compress the fixed regulatory review clocks, and every generated document must still pass human medical, statistical, and regulatory review before use. The build is an independent research artifact, not an active IND or IDE and not regulatory advice.

5.2 Future Work

Future work extends the method along three axes. First, coverage: applying the same single-prompt pipeline to the Phase 2-to-3 briefing package and the Phase 3 protocol [5], and to the cohort-review and clinical-hold response packages under their real clocks. Second, integration: connecting the build to live electronic data capture and electronic document management feeds so that tables are generated directly from validated data, and adding automated verification of generated documents in the spirit of the verification-before-generation standard [13]. Third, breadth: extending beyond PDAC to glioblastoma and lung adenocarcinoma [24, 19], and connecting to the national platform and site-documentation pipelines so that a site can produce its full document set on demand [2]. Each extension preserves the same discipline: grounded figures, name-matched files, real-time monitorability, human in the loop.

6 Conclusions

A single repository based LLM prompt, which first writes and then executes a schedule of sub-prompts through a mermaid to draft to full to final pipeline, can generate every relevant large Phase 1 document and thereby hasten the entire Phase 1 process. The acceleration is specific and honest: it compresses the administrative and preparation bucket, where document white space between phases is lost, and leaves the clinical-operational and fixed regulatory-review buckets untouched. Applied to the six highest-value document targets, the method turns iterations that took weeks to months into iterations that now take one to four days.

The method is monitorable and verifiable rather than merely fast. Its 24 figures ground the prose and reproduce identically in LaTeX; its repository, directory, and file names match the artifacts on disk; and every file is committed to GitHub in real time. Because each probable risk is reduced to a low residual by a concrete control, probable benefit exceeds probable risk for enrolled PDAC patients who cannot wait, and faster, safer document administration translates, through earlier treatment, into extended progression-free and overall survival.

This application is credible because the LLM trust it depends on was established incrementally: the 2024 to 2026 application papers, the National Platform framework [2], and the prior single-prompt multi-file builds of the Phase 1 and Phase 2 daraxonrasib PDAC protocols and the law and bill repositories [4, 5, 6, 11, 3, 12, 13, 14] together show that the same single-prompt approach, now turned on the document-generation problem itself, is a natural and well-supported next step.

7 References and Back Matter

The reference list renders below through the `ieeetr` style from `references.bib`. Every reference carries a clickable URL and a clickable DOI URL where relevant, and long links break on any character so none runs off the right margin. The five DARAXONRASIB entries, the three main documents (the 2030 60-second PDAC simulation, the H. R. 9510 bill v5.0, and the National Platform framework), the Phase 1 and Phase 2 protocols, the cited single-prompt repositories, the 2025 application papers, and the four research sources are all included.

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Ethical Disclosures

This is an independent research paper and practical adoption guide. No real patients were enrolled, no protected health information was used, and no IRB approval was sought or obtained. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference. This document is not an active IND or IDE, is not an approved protocol, and is not medical or regulatory advice; it is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor.

Data Availability

All paper sources (the `mermaid`, `draft-paper`, `full-paper`, and `final-paper` stages) are available in the public GitHub repository and archived on Zenodo under the Creative Commons Attribution 4.0 International (CC BY 4.0) license at DOI 10.5281/zenodo.21018646.

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